The general mathematics of the Laplace transform is very similar to that of the Fourier transform. In both cases, predetermined waveforms are multiplied by the time domain signal, and the result integrated. At first glance, it would appear that the strategy of the Laplace transform is the same as the Fourier transform: correlate the time domain signal with a set of basis functions to decompose the waveform. Not true! Even though the mathematics is much the same, the rationale behind the two techniques is very different. The Laplace transform *probes* the time domain waveform to identify its key features: the *frequencies* of the sinusoids, and the *decay constants* of the exponentials. An example will show how this works.

The center column in Fig. 32-5 shows the impulse response of the *RLC* notch filter discussed in Chapter 30. It contains an impulse at $t = 0$, followed by an exponentially decaying sinusoid. As illustrated in (a) through (e), we will *probe* this impulse response with various exponentially decaying sinusoids. Each of these probing waveforms is characterized by two parameters: ω, that determines the sinusoidal frequency, and σ, that determines the decay rate. In other words, each probing waveform corresponds to a different location in the s-plane, as shown by the s-plane diagram in Fig. 32-4. The impulse response is probed by *multiplying* it with these waveforms, and then integrating the result from $t = -\infty$ to $+\infty$. This action is shown in the right column. Our goal is to find combinations of σ and ω that exactly *cancel* the impulse response being investigated. This cancellation can occur in two forms: the area under the curve can be either *zero*, or just *barely infinite*. All other results are uninteresting and can be ignored. Locations in the s-plane that produce a zero cancellation are called **zeros** of the system. Likewise, locations that produce the "just barely infinite" type of cancellation are called **poles**. Poles and zeros are analogous to the mountains and valleys in our train story, representing the terrain "around" the frequency response.

To start, consider what happens when the probing waveform decreases in amplitude as time advances, as shown in (a). This will occur whenever $\sigma > 0$ (the right half of the s-plane). Since both the impulse response and the probe becomes smaller with increasing time, the product of the two will also have this same characteristic. When the product of the two waveforms is integrated from negative to positive infinity, the result will be some number that is not especially interesting. In particular, a decreasing probe

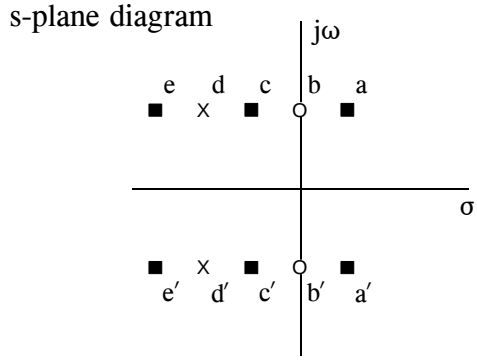


FIGURE 32-4
Pole-zero example. The notch filter has two poles (represented by ×) and two zeros (represented by ○). This s-plane diagram shows the five locations we will "probe" in this example to analyze this system. (Figure 30-5 is a continuation of this example).